A Multi-Architecture Neural Operator Digital Twin: A Proof-of-Concept Surrogate Methodology for Cold-Leg LOCA Screening in AP1000 Nuclear Reactors

Punyansh Thakur, Anushka Paul, Ayush Kumar Bhadra, and Vinay Kumar

***Abstract*—High-fidelity computational-fluid-dynamics (CFD) analysis is generally too costly for repeated online accident-screening studies. This paper presents a proof-of-concept AP1000 digital-twin methodology that couples spatial surrogate modeling to a plant-indicator classifier. Three neural operators map inlet velocity, prescribed break-severity proxy, and coolant temperature to pressure, velocity magnitude, turbulent kinetic energy, and temperature on a shared 25,000-point mesh: DeepONetFourier (1,451,012 parameters), a Transolver-inspired token operator (3,244,872), and a grade-structured Clifford-algebra operator (37,380). The evaluated benchmark comprises 2,000 physics-inspired analytic synthetic cases, including 300 held-out cases. The archived DeepONetFourier checkpoint achieved pooled R^2 values of 0.9961, 0.9958, 0.9951, and 0.9962 for the four normalized fields, respectively; a separate seed-42 full-test run of the Transolver-inspired operator achieved 0.9918, 0.9932, 0.9806, and 0.9951. A dataset-matched unmodified DeepONet baseline is also reported; cross-architecture accuracy ranking is avoided because the objectives are not identical and operator stability has not yet been established. Separately, a gradient-boosting classifier trained on open PCTRAN-simulated NPPAD records with synthetic transition augmentation achieved ROC-AUC 0.990554 and recall 0.989063 on one fixed row-level split. A translation-path diagnostic confirms executable coupling, but the label-conditioned break input prevents interpretation as independent accident diagnosis. The results establish reproducible workflow feasibility under the analytic benchmark. Full-fidelity CFD or experimental validation, label-independent and group-separated detection tests, matched-loss multi-seed comparisons, and reliability-oriented operating-point analysis remain necessary before safety-relevant assessment.**

***Index Terms*—Digital Twin, DeepONet, Transolver, Clifford Neural Operator, Geometric Algebra, Transformer Neural Operator, LOCA, AP1000, Nuclear Safety, Surrogate Modeling, Fourier Feature Encoding**

I. INTRODUCTION

The AP1000 is a Generation III+ pressurized-water reactor designed by Westinghouse with passive safety features. A cold-leg loss-of-coolant accident (LOCA) is a postulated rupture in the primary coolant return piping whose safety analysis requires system-level thermal-hydraulic modeling, uncertainty treatment, and comparison with applicable evidence [10], [11].

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This study investigates data-driven reconstruction and screening as a research supplement; it does not replace engineered protection, qualified analysis codes, or operator authority.

High-fidelity CFD can predict spatial thermohydraulic states, but its computational cost limits repeated online evaluation. Neural operators—machine-learning models that learn mappings between function spaces—have emerged as PDE surrogates. Lu et al. [1] introduced Deep Operator Networks (DeepONet), decomposing the solution operator $\mathcal{G} : \mathcal{U} \to \mathcal{V}$ via a branch–trunk factorization. The coordinate MLP in the present baseline is susceptible to spectral bias [7] and uses fixed ReLU activations. The studied extension adds Fourier coordinates, adaptive activations, and serialized-node smoothness proxies; Section V-A defines their scope explicitly.

Subsequent work has explored alternative neural operator paradigms: Transolver [5] applies transformer attention to mesh-compressed tokens for general-geometry PDE solving, and Clifford Neural Operators [6] use geometric-algebra representations to encode grade structure. Recent nuclear applications include DeepONet-based hot-leg virtual sensing and DeepONet/FNO surrogates for a helical-coil steam generator [15], [16]. To the authors' knowledge, no prior report jointly evaluates Transolver- and Clifford-style operators in a cold-leg LOCA field-surrogate-to-classifier workflow; this novelty claim is limited to that specific combination and to the documented public-source search.

The Fourier Neural Operator (FNO) [4] is a natural comparison candidate. The present study instead focuses on branch–trunk, token-attention, and Clifford-algebra parameterizations under one fixed point-cloud representation. An FNO benchmark with a documented gridding/interpolation protocol is reserved for a controlled follow-up.

The main contributions of this paper are summarized as follows:

1) We design and implement three neural-operator architectures—an upgraded DeepONetFourier, a Transolver-inspired token operator, and a Clifford-algebra operator—within a unified proof-of-concept pipeline for LOCA screening.
2) We introduce four changes to baseline DeepONet—random Fourier feature encoding ($\sigma = 10.0$, $m = 256$), per-neuron adaptive GELU activations, an index-gradient matching term ($\beta = 0.1$), and a velocity-variation term ($\lambda = 0.01$)— reducing the measured

parameter count from 3,162,116 to 1,451,012 (54.1%) under the synthetic benchmark. The latter two terms are serialized-node regularizers rather than geometry-aware physics derivatives.
3) We evaluate the Transolver-inspired operator (mesh-token attention with $M = 64$ tokens, 3,244,872 parameters) and the grade-structured Clifford-algebra operator (37,380 parameters) as field surrogates. Novelty is stated only relative to the targeted search above, and the dense Clifford readout precludes an exact end-to-end equivariance guarantee.
4) We provide an artifact-backed comparison along the dimensions supported uniformly across the three implementations: architecture, parameter count, objective, output interface, and matched-scale qualitative reconstruction. Quantitative accuracy is reported where a complete evaluation artifact is available.
5) We couple the surrogate models to a LOCA indicator-translation demonstration. The archived single-split classifier reaches ROC-AUC 0.990554 on PCTRAN-simulated NPPAD data with synthetic transition augmentation. A five-seed translation-path ablation is included with its label-conditioned-input limitation made explicit.

The remainder of this paper is organized as follows. Section II reviews related work on neural operators and machine learning for nuclear power plant safety. Section III formulates the surrogate-modeling problem and its input-output specification. Section IV describes the end-to-end system architecture and the five-stage digital twin pipeline. Section V details the three neural operator architectures and their respective improvements over baseline DeepONet. Section VI presents the downstream LOCA indicator detection pipeline, including the NPPAD feature mapping and severity model. Section VII describes implementation details, software, hardware, and the training protocol. Section VIII reports experimental results under the synthetic benchmark. Section IX discusses the methodological contributions, architectural trade-offs, and the limitations that define the path to full-fidelity validation. Section X concludes the paper.

II. Related Work

A. Neural Operators and DeepONet

Neural operators learn mappings between function spaces [3]. Two widely used families are the Fourier Neural Operator (FNO) [4], whose standard FFT realization uses a regular grid, and DeepONet [1], [2], which factorizes the learned mapping into branch and trunk networks and can query arbitrary coordinates. Coordinate MLPs can preferentially fit low-frequency components [7]; Fourier feature encoding [8] and derivative-aware training [9] are two relevant remedies.

B. Transformer and Geometric Neural Operators

Transolver [5] compresses N-node meshes to $M \ll N$ learned tokens via soft-assignment attention, achieving $\mathcal{O}(M^2)$ token-attention complexity; GNOT [14] applies a related transformer-based approach to operator learning. Clifford Neural Operators [6] take a different route by embedding physical fields in Clifford algebra. Equivariance depends on maintaining grade-respecting operations through the output map; the implementation studied here uses a dense grade-mixing readout, so it is described as grade-structured rather than exactly equivariant. The domain-novelty statement is limited to the targeted-search scope stated in Section I.

C. Machine Learning for Nuclear Power Plant Safety

Machine-learning research for nuclear reactors spans transient classification, fault diagnosis, sensor monitoring, and thermal-hydraulic surrogates, while important barriers include limited representative data, distribution shift, robustness, and interpretability [13]. Digital-twin concepts emphasize a maintained connection between physical and digital assets [12]; recent nuclear virtual-sensing work illustrates this direction with neural operators [15]. The present work connects analytic spatial-field surrogates to an ML classifier in a unified proof-of-concept workflow and, within the targeted-search scope above, compares multiple operator paradigms for cold-leg LOCA screening.

D. Reliability-Engineering Perspective on AI-Based Detection

Reliability-oriented fault detection is evaluated through an operating characteristic, not a single accuracy number. Probability of detection and false-alarm rate must be reported as functions of the threshold and operating regime because their consequences are asymmetric [18]. The single threshold used here is therefore descriptive; it is not a justified alarm setting.

For safety-relevant software, IEEE Std 1012 defines lifecycle V&V processes, while recent nuclear guidance emphasizes trustworthy design, data governance, testing, monitoring, human oversight, and defense in depth for AI applications [19]–[21]. These sources motivate a staged safety case comprising requirements traceability, independent test evidence, uncertainty and distribution-shift analysis, configuration control, and documented human authority. The present prototype supplies none of the evidence needed for certification and is not proposed for plant protection or autonomous actuation.

Validation maturity must also be separated by model class. Nuclear thermal-hydraulic system codes such as TRACE and RELAP5 have long-established assessment practices against qualified experimental databases and plant transients; best-estimate-code qualification explicitly addresses code, user, nodalization, and uncertainty effects [22], [23]. In contrast, this work evaluates spatial surrogates only against an analytic mock generator. Its contribution is architectural and workflow-level feasibility, not parity with a qualified system-code analysis.

III. Problem Formulation

Let $\Omega \subset \mathbb{R}^3$ denote the evaluated straight cylindrical cold-leg proxy segment, with diameter $D = 0.7\,\mathrm{m}$ and length $10D$. The analytic benchmark does not include the elbow geometry present in the planned Fluent model. Define the parameter vector $\boldsymbol{u} = [v, b, T]^\top \in \mathbb{R}^3$, where $v \in [4.0, 6.0]\,\mathrm{m/s}$ is the inlet velocity, $b \in [0, 10]\%$ is the prescribed break-severity proxy, and $T \in [290, 320]$°C is the coolant temperature. In the evaluated generator, b modulates closed-form fields; no physical opening or break-area mesh is resolved.

The evaluated goal is to learn the parametric mapping defined by the analytic generator; future CFD validation would replace these targets with independently verified solver fields:

$$\mathcal{G}_\theta(\boldsymbol{u})(\boldsymbol{y}) = \hat{\boldsymbol{s}}(\boldsymbol{y}) = [\hat{p},\ |\hat{\boldsymbol{v}}|,\ \hat{k},\ \hat{\tilde{T}}]^\top \in \mathbb{R}^4 \tag{1}$$

for every spatial location $\boldsymbol{y} \in \Omega$, evaluated simultaneously across $N = 25{,}000$ mesh nodes: output $\hat{\boldsymbol{S}} \in \mathbb{R}^{B\times 4\times N}$. The stored dataset and model output contain these four scalar fields only; no (v_x, v_y, v_z) tensor is produced. Consequently, the implemented velocity regularizer cannot represent $\nabla \cdot \boldsymbol{v}$ or establish mass conservation. Section V-A therefore defines it as a serialized-node velocity-variation proxy and identifies geometry-aware vector conservation as future work.

A. Analytic Synthetic-Benchmark Setup

The evaluated generator assumes constant coolant density of 720 kg/m^3 and a 15.5 MPa pressure scale. It does not solve the RANS equations. Instead, closed-form axial, radial, and break-localized expressions generate pressure, velocity magnitude, turbulent kinetic energy, and temperature on the straight cylindrical proxy mesh, with small stochastic perturbations. The dataset comprises 2,000 parameter combinations: 20 velocity $\times$ 10 break-size $\times$ 10 temperature levels, split deterministically with seed 42 into training (1,400), validation (300), and test (300) samples.

Accordingly, the present benchmark should be interpreted as a parametric field-regression proxy with operator-learning structure rather than a fully validated PDE-solution benchmark.

IV. System Architecture and Pipeline

The digital twin operates as a five-stage cascade pipeline (Fig. 1), with each stage transforming data from one physical or learned representation into the next, culminating in a LOCA classifier score for screening research. A master controller orchestrates the cascade and supports runtime selection of any neural operator architecture through a unified configuration interface.

A. Stage 1: Parametric Field-Data Generation

The training data originates from a parametric sweep of the three-dimensional input space (v, b, T). Two data generation pathways are supported:

Pathway A — ANSYS Fluent Automation (Unevaluated Prototype): The repository includes a journal-generation prototype that can configure a steady pressure-based solver, RNG k–ε closure, inlet/outlet conditions, a pre-existing break boundary zone, 1,000 solver iterations, and CSV field export. This pathway did not produce the dataset or results reported in this paper. Before it can serve as a validation path, the geometry and break-area parameterization, mesh independence, solver convergence, and export consistency must be verified in Fluent.

Pathway B — Analytic Synthetic Generator (Evaluated): A synthetic data generator produces fields on a shared 25,000-node mesh (cylindrical pipe, diameter $D = 0.7$ m, length $10D$) incorporating:

- **Pressure**: Linear axial term ($\Delta p = 50{,}000 \cdot v/5$ Pa) + localized Gaussian break-proxy drop ($200{,}000 \cdot b_{\text{norm}}$ Pa, $\sigma = 0.8$ m) + prescribed radial shape term ($5{,}000(1-r^2)$ Pa)
- **Velocity**: Parabolic profile $v(r) = v_{\text{in}}(1 - r^\alpha)$, with $\alpha = 2 - 0.8 b_{\text{norm}} e^{-\frac{(x-x_{\text{break}})^2}{2\sigma^2}}$ (flattens near break) and 30% reduction at full break
- **Turbulence**: Base $k = 0.01v^2$, wall-layer increase (factor $1+1.5r$), break-induced spike ($5b_{\text{norm}}$, localized at break)
- **Temperature**: Kelvin base + radial gradient ($3(1-r^2)$ K) + downstream break hot-spot ($15b_{\text{norm}}$ K) + axial cooling (2 K over pipe length)

where $b_{\text{norm}} = b/10 \in [0, 1]$ and $r = \sqrt{y^2 + z^2}/(D/2)$ is the normalized radial position. The fields include small Gaussian or uniform stochastic perturbations specified by the generator; these perturbations are heuristic and are not a turbulence model.

The 2,000 parameter combinations form a $20 \times 10 \times 10$ structured grid: 20 velocity levels in $[4.0, 6.0]$ m/s, 10 break sizes in $[0, 10]\%$, and 10 temperatures in $[290, 320]$°C.

B. Stage 2: Preprocessing and Normalization

The `DeepONetDataPreprocessor` module converts raw CSV files into a training-ready HDF5 dataset:

1) **Loading**: Each CSV is parsed into a DataFrame, source headers are mapped to consistent internal field names, and mesh consistency is checked across all simulations ($N = 25{,}000$ nodes).
2) **Tensor construction**:

$$\boldsymbol{U} \in \mathbb{R}^{2000\times 3} \qquad \text{(branch inputs: } v, b, T\text{)}, \tag{2}$$
$$\boldsymbol{Y} \in \mathbb{R}^{N\times 3} \qquad \text{(trunk inputs: } x, y, z\text{)}, \tag{3}$$
$$\boldsymbol{S} \in \mathbb{R}^{2000\times 4\times N} \qquad \text{(target fields)}. \tag{4}$$

3) **Normalization**: Per-field MinMaxScaler to $[0, 1]$. Branch inputs are scaled jointly; trunk coordinates are scaled jointly; each target field (pressure, velocity, turbulence, temperature) receives its own independent scaler. All fitted scalers are serialized (pickled) for use during inference denormalization.
4) **Splitting**: Deterministic random 70/15/15% split into training (1,400), validation (300), and test (300) samples using seed 42. No label stratification is applied. The trunk coordinates $\boldsymbol{Y}$ are shared across all splits (same mesh for all simulations).
5) **Storage**: The splits are saved to a single HDF5 file (`deeponet_dataset.h5`) with groups `train/`, `val/`, `test/`, each containing datasets `branch`, `trunk`, `targets`.

Stage 1: Data Generation

Evaluated analytic generator $\longrightarrow$ 2,000 cases $\times$ 25,000 nodes $\times$ (3 coordinates + 4 retained fields)

$\downarrow$

Stage 2: Preprocessing & Normalization

CSV $\rightarrow$ HDF5: Branch $\boldsymbol{U} \in \mathbb{R}^{2000\times3}$, Trunk $\boldsymbol{Y} \in \mathbb{R}^{25000\times3}$, Targets $\boldsymbol{S} \in \mathbb{R}^{2000\times4\times25000}$ | MinMaxScaler per field

$\downarrow$

Stage 3: Neural Operator Training

DeepONetFourier $\xrightarrow{\text{MSE + serialized-node proxies}}$ `deeponet_fourier_best.pth`; token/Clifford operators $\xrightarrow{\text{MSE}}$ architecture-specific checkpoints.

$\downarrow$

Stage 4: LOCA Indicator Classifier Training

Open PCTRAN-simulated NPPAD data (302 Normal + 45,176 LOCA rows) + 3,800 synthetic transition samples $\rightarrow$ GBC/MLP $\rightarrow$ `loca_detector.pkl`

$\downarrow$

Stage 5: Inference Pipeline (Screening Mode)

$\boldsymbol{u} \rightarrow$ Normalize $\rightarrow$ Neural Operator $\rightarrow$ Denormalize $\rightarrow$ Feature Extraction $\rightarrow$ NPPAD Mapping $\rightarrow$ LOCA Classifier $\rightarrow$ S_{LOCA}

Single-case timing path available; reproducible matched-hardware benchmark deferred.

Fig. 1. End-to-end digital-twin proof-of-concept pipeline. Stages 1–4 run offline and Stage 5 executes the screening demonstration. The evaluated field data are synthetic, NPPAD records are PCTRAN-simulated, and this workflow is neither a plant protection system nor an operationally validated digital twin.

C. Stage 3: Neural Operator Training

The training stage supports three interchangeable neural operator architectures (detailed in Section V). The architecture is selected at runtime via a `model_config.yaml` configuration file, with all operators sharing the same data loader and evaluation metrics.

Data loading: A PyTorch `Dataset` reads HDF5 lazily. Each sample is a tuple (branch $\boldsymbol{u}_i \in \mathbb{R}^3$, trunk $\boldsymbol{Y} \in \mathbb{R}^{N\times3}$, targets $\boldsymbol{s}_i \in \mathbb{R}^{4\times N}$). The archived DeepONet and DeepONet-Fourier runs use $B = 4$; the Tier-2 operator configuration uses $B = 16$.

Training loop: For each batch:

1) Forward pass: $\hat{\boldsymbol{S}} = \mathcal{G}_\theta(\boldsymbol{U}_{\text{batch}}, \boldsymbol{Y}) \in \mathbb{R}^{B\times4\times N}$
2) Loss computation: In the executed DeepONetFourier path, the base MSE is added to a "Sobolev" module that itself contains an MSE term plus a central-difference term along stored node index. A second module applies the same index difference to velocity magnitude. Thus, with D_{idx} denoting that serialized-node difference, the effective objective is
$$\mathcal{L}_{\text{DF}} = 2\,\mathcal{L}_{\text{MSE}} + \beta \left\| D_{\text{idx}}\hat{\boldsymbol{S}} - D_{\text{idx}}\boldsymbol{S} \right\|_F^2 + \lambda \left\| D_{\text{idx}} |\hat{\boldsymbol{v}}| \right\|_F^2, \tag{5}$$
where $\beta = 0.1$ and $\lambda = 0.01$. Because the mesh rows are not ordered as a physical one-dimensional stencil, the latter terms are smoothness proxies, not $\nabla\boldsymbol{S}$ and $\nabla\cdot\boldsymbol{v}$. The Transolver-inspired and Clifford-algebra operators use MSE only.
3) Backward pass with Automatic Mixed Precision (AMP) arithmetic (FP16 forward, FP32 gradients)
4) Adam optimizer step with gradient clipping ($\|\nabla\theta\|_{\max} = 1.0$)

Scheduling: ReduceLROnPlateau (patience 20, factor 0.5) for DeepONetFourier; CosineAnnealingLR for Transolver and Clifford. Early stopping with patience 50 monitors validation loss.

Evaluation metrics: Per-field R^2, relative L_2 error, and MAE are computed on validation/test data; the exact pooled test definitions used in Table VII are stated there. The evaluated checkpoints are architecture-specific: `deeponet_fourier_best.pth`, `transolver_best.pth`, and `clifford_best.pth`.

D. Stage 4: LOCA Indicator Classifier Training

The `LOCADetector` module trains a supervised binary classifier completely independently from the neural operator.

Data source: The open Nuclear Power Plant Accident Database (NPPAD), generated with the PCTRAN simulator rather than measured at an operating plant [17], is loaded from the `data/nppad/operation_csv_data/` directory:

- `Normal/` — 1 CSV, 302 timestep rows (steady-state operation)
- `LOCA/` — 100 CSVs, ~45,176 timestep rows (accident transients)

Feature extraction: Seven NPPAD signals are selected from each row: primary pressure P, average temperature T_{avg}, RCS flow W_{RCA}, SG-A pressure P_{SGA}, subcooling margin ΔT_{sub}, Departure from Nucleate Boiling Ratio (DNBR), and hot-cold leg temperature differential $\Delta T_{\text{HL-CL}} = T_{\text{HA}} - T_{\text{CA}}$.

Transitional data augmentation: To avoid an exclusively endpoint-like training distribution, 19 severity levels $s \in \{0.05, 0.10, \ldots, 0.95\}$ are generated with 200 samples each. At each level, feature values are interpolated between Normal and LOCA distributions, with labels drawn as Bernoulli(s). This adds 3,800 synthetic samples, giving 49,278 rows before the split. This construction does not establish probability calibration.

Classifier: A Gradient Boosting Classifier (200 trees, depth 4, learning rate 0.05, min. 20 samples/leaf) trained on StandardScaler-normalized features. An 80/20 stratified row-level split with seed 42 is used for training and evaluation. It is not grouped by source transient; consequently, the reported score is a within-dataset classifier result rather than an unseen-transient validation.

Artifacts: The trained model, scaler, and metrics are serialized to `loca_detector.pkl`.

E. Stage 5: Inference Pipeline (Screening Mode)

The `DigitalTwinInference` class orchestrates the forward pass through the complete digital twin at runtime. The five steps execute sequentially within each inference call:

Step 1 — Input normalization: Physical parameters (v, b, T) are transformed using the branch MinMaxScaler from Stage 2:

$$\boldsymbol{u}_{\text{norm}} = \frac{\boldsymbol{u} - \boldsymbol{u}_{\min}}{\boldsymbol{u}_{\max} - \boldsymbol{u}_{\min}} \in [0, 1]^3 \tag{6}$$

Step 2 — Neural operator forward pass: The normalized branch input and pre-stored normalized trunk coordinates are fed to the loaded neural operator:

$$\hat{\boldsymbol{S}}_{\text{norm}} = \mathcal{G}_\theta(\boldsymbol{u}_{\text{norm}}, \boldsymbol{Y}_{\text{norm}}) \in \mathbb{R}^{1\times 4\times N} \tag{7}$$

This step runs within a `torch.no_grad()` context on the selected device (CUDA when available, otherwise CPU). Reproducible latency values are not inferred from the source code and are treated separately in Section VIII-E.

Step 3 — Field denormalization: Each predicted field is inverse-transformed using its per-field MinMaxScaler:

$$\hat{s}_i(\boldsymbol{y}) = \hat{s}_{i,\text{norm}} \cdot (s_{i,\max} - s_{i,\min}) + s_{i,\min} \tag{8}$$

producing physical-unit fields (Pa, m/s, m^2/s^2, K).

Step 4 — Feature extraction and NPPAD mapping: The `FeatureTranslator` computes 11 predicted-field-derived scalar features from the denormalized outputs:

- Pressure: spatial mean, inlet/outlet-decile drop, that drop divided by axial span, and standard deviation
- Velocity: mean at inlet plane ($x < x_{10\%}$), standard deviation
- Turbulence: peak k, spatial mean k
- Temperature: spatial mean, max-min difference
- Mass flow rate: $\dot{m} = \rho \cdot \bar{v}_{\text{inlet}} \cdot A_{\text{pipe}}$

These local field-derived features are then combined with the original input parameters through the blended severity model (Eq. 29) to produce seven NPPAD-like mapped signals. The severity model uses a sigmoid curve centered at $b = 5\%$ and blends input-derived severity (80%) with field-derived anomaly signals (20%). Because the prescribed break size dominates this mapping, the main result cannot isolate an independent contribution from reconstructed fields.

Step 5 — LOCA indicator classification: The seven NPPAD signals are cast to a named DataFrame (matching the training feature order), scaled by the stored StandardScaler, and passed to the Gradient Boosting Classifier:

$$\boldsymbol{x}_{\text{NPPAD}} = [P,\, T_{\text{avg}},\, W_{\text{RCA}},\, P_{\text{SGA}},\, \Delta T_{\text{sub}},\, \text{DNBR},\, \Delta T_{\text{HL-CL}}] \tag{9}$$

$$S_{\text{LOCA}} = \text{GBC}_\theta(\text{StandardScaler}(\boldsymbol{x}_{\text{NPPAD}})) \tag{10}$$

A threshold of 0.5 converts the classifier score S_{LOCA} to a binary decision. The score has not been independently calibrated as a probability or uncertainty bound.

F. Time-Series and Benchmarking Modes

Beyond single-shot inference, the pipeline supports:

Time-series demonstration: A prescribed sequence of (v, b, T) tuples can emulate an evolving synthetic scenario (e.g., break size grows linearly from 0% to 10% over 40 s). Repeated steady-state inferences produce a classifier-score trajectory for analysis. This loop is not a learned transient model and is not evaluated for operator decision support.

Benchmark mode: The command-line path can repeat model inference and summarize elapsed time. The present artifacts do not preserve a matched-hardware benchmark, so no CFD speedup is treated as an experimental result.

V. Neural Operator Architectures

We implement three neural operator architectures organized in two tiers, each addressing distinct limitations of the original DeepONet baseline. Table I provides a preliminary comparison under the present synthetic benchmark. An unmodified DeepONet was subsequently trained on the same 70/15/15 split with batch size 4, learning rate 10^{-3}, and CUDA. Its archived run stopped after 119 epochs with 3,162,116 parameters, test MSE 1.97×10^{-4}, and minimum per-field test $R^2 = 0.99332$. This establishes a dataset-matched baseline, but the comparison remains non-identical: DeepONetFourier uses the additional index-space terms in Eq. (5), whereas the baseline, Transolver-inspired, and Clifford-algebra models use MSE-only objectives.

A. Tier 1: DeepONetFourier — Upgraded Baseline

The baseline DeepONet approximation [1], [2] decomposes the operator as:

$$\mathcal{G}(\boldsymbol{u})(\boldsymbol{y}) \approx \sum_{k=1}^{p} b_k(\boldsymbol{u}) \cdot t_k(\boldsymbol{y}) + b_0 \tag{11}$$

where $\{b_k\}$ are basis coefficients from the branch network $\mathcal{B} : \mathbb{R}^3 \to \mathbb{R}^p$ and $\{t_k\}$ are basis functions from the trunk network $\mathcal{T} : \mathbb{R}^3 \to \mathbb{R}^p$, with $p = 256$.

We introduce four changes that reduce the measured parameter count by 54.1% relative to the retrained baseline and improve representational flexibility under the synthetic benchmark:

1) Improvement 1: Random Fourier Feature Encoding: Standard MLPs suffer from *spectral bias*: they fit low-frequency components first and slowly learn sharp spatial gradients near walls and turbulence interfaces [7]. We mitigate this tendency by replacing raw coordinate input with Fourier feature encoding [8]:

$$\gamma(\boldsymbol{y}) = [\sin(2\pi \boldsymbol{B}\boldsymbol{y}),\ \cos(2\pi \boldsymbol{B}\boldsymbol{y})] \in \mathbb{R}^{2m} \tag{12}$$

where $\boldsymbol{B} \in \mathbb{R}^{3\times m}$ is a *fixed* random frequency matrix sampled from $\mathcal{N}(0, \sigma^2)$ with $m = 256$, $\sigma = 10.0$, yielding

TABLE I
ARCHITECTURE AND EVIDENCE COMPARISON ON THE SYNTHETIC BENCHMARK. DEEPONETFOURIER USES ADDITIONAL SERIALIZED-NODE PROXY TERMS; THE OTHER MODELS USE MSE ONLY. ALL EVALUATED FIELDS SHARE ONE FIXED MESH, SO GEOMETRY-TRANSFER CLAIMS ARE NOT TESTED.

Criterion	**Baseline DeepONet**	**DeepONetFourier (Ours)**	**Transolver-inspired**	**Clifford-algebra**
Parameters	3,162,116	1,451,012	3,244,872	37,380
Frequency mechanism	Raw coordinates	Fourier encoding	Learned tokenization	Grade-structured product
Global spatial context	Local (dot product)	Local (dot product)	Global (attention)	Local
Output symmetry	None	None	None	Not guaranteed (dense readout)
Training objective	MSE	MSE + index-space proxies	MSE	MSE
Held-out evidence retained in this study	Four-field metrics	Four-field metrics	Matched-scale case-0 panel	Matched-scale case-0 panel

a 512-dimensional input to the trunk network. The fixed random projection simultaneously injects sinusoidal features at many spatial frequencies, enabling the trunk to construct both slowly-varying bulk-flow patterns and rapidly-varying boundary-layer profiles from the first layer onward.

2) Improvement 2: Per-Neuron Adaptive GELU Activation: Each neuron in the trunk's first two hidden layers employs a *learnable slope* parameter a_i (initialized to 1.0):

$$f_i(x) = \mathrm{GELU}(a_i \cdot x) \tag{13}$$

The parameter a_i is optimized via backpropagation independently for each neuron, allowing the effective nonlinearity curvature to adapt to the local spatial mapping landscape. This replaces the fixed ReLU activations of the baseline, which cannot adapt steepness during training.

3) Improvement 3: Serialized-Node Gradient-Matching Proxy: The executed training path sets `use_autograd=False`; both predicted and target differences are computed by a central stencil along the stored node axis. The implemented module, inspired by Sobolev training [9], is therefore

$$\mathcal{L}_{\text{idx}} = \mathcal{L}_{\text{MSE}} + \beta \left\| D_{\text{idx}}\hat{\boldsymbol{S}} - D_{\text{idx}}\boldsymbol{S} \right\|_F^2, \tag{14}$$

with $\beta = 0.1$. Since node rows are not a geometry-aware stencil, D_{idx} has no consistent physical direction or length scale. It regularizes variation in serialization order but cannot substantiate claims about pressure-gradient, boundary-layer, or derivative fidelity. Geometry-aware Green–Gauss, local least-squares, or validated nearest-neighbor reconstruction, followed by derivative-error evaluation, is reserved for the full-fidelity study.

4) Improvement 4: Serialized-Node Velocity-Variation Proxy: The implementation receives only scalar velocity magnitude and applies a central difference along stored node index:

$$\mathcal{L}_{\text{var}} = \lambda \left\| D_{\text{idx}} |\hat{\boldsymbol{v}}| \right\|_F^2, \tag{15}$$

with $\lambda = 0.01$. This discourages rapid index-to-index changes but is neither scalar-field divergence nor vector divergence. The actual total objective is Eq. (5); no mass-conservation conclusion is drawn.

5) Architecture Summary: The complete architecture uses four independent branch-trunk pairs (one per output field):

TABLE II
DEEPONETFOURIER ARCHITECTURE VS. THE DATASET-MATCHED RETRAINED DEEPONET BASELINE

Component	**Baseline**	**Our Upgrade**
Branch	3→256→512→512→256 ReLU + Drop(10%)	3→128→256→256 GELU + Drop(5%)
Trunk input Trunk hidden	Raw (x, y, z) 3→256→512→256 ReLU + Drop(10%)	Fourier(3→512) 512→256→256 Adaptive GELU
Loss Parameters	MSE 3,162,116	MSE + index-space proxies 1,451,012

B. *Tier 2: Transolver-Inspired Token Operator*

For the shared fixed mesh, DeepONetFourier forms predictions through branch–trunk inner products at each query coordinate. As a complementary Tier-2 parameterization, the custom token operator introduces global interactions among learned tokens using a Transolver-inspired compression-and-attention pattern [5]. It is not claimed to reproduce every component or benchmark protocol of the published Transolver.

1) Mesh Tokenization via Soft Assignment: The N-node mesh is compressed into $M = 64$ learned tokens via a learnable soft-assignment:

$$\boldsymbol{f}_n = \boldsymbol{W}_{\text{proj}}\, \boldsymbol{y}_n \in \mathbb{R}^d \quad \text{(node embedding)} \tag{16}$$

$$\alpha_{nm} = \frac{\exp(\boldsymbol{f}_n^\top \boldsymbol{q}_m)}{\sum_{m'} \exp(\boldsymbol{f}_n^\top \boldsymbol{q}_{m'})} \quad \text{(soft assignment)} \tag{17}$$

$$\boldsymbol{\tau}_m = \sum_{n=1}^{N} \alpha_{nm}\, \boldsymbol{f}_n \in \mathbb{R}^d \quad \text{(token aggregation)} \tag{18}$$

where $\boldsymbol{W}_{\text{proj}} \in \mathbb{R}^{d\times 3}$, $\boldsymbol{q}_m$ are learned query vectors, and $d = 256$ is the embedding dimension. The assignment weights α_{nm} are learned end-to-end and provide a data-adaptive mesh compression; no claim is made that individual tokens correspond to named physical regions without a token-interpretability analysis.

2) Branch Conditioning: Physics parameters $\boldsymbol{u}$ are encoded by a branch encoder ($3 \to 256 \to 256$, GELU) and added to all M tokens:

$$\boldsymbol{\tau}'_m = \boldsymbol{\tau}_m + \mathcal{B}(\boldsymbol{u}) \tag{19}$$

3) Self-Attention Over Tokens: Four transformer blocks with 8-head self-attention (head dimension 32) process the conditioned tokens:

$$\mathrm{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{20}$$

$$\boldsymbol{x} \leftarrow \boldsymbol{x} + \mathrm{Dropout}(\mathrm{MHSA}(\mathrm{LN}(\boldsymbol{x}))) \tag{21}$$

$$\boldsymbol{x} \leftarrow \boldsymbol{x} + \mathrm{Dropout}(\mathrm{MLP}(\mathrm{LN}(\boldsymbol{x}))) \tag{22}$$

The self-attention substep has $\mathcal{O}(M^2) = \mathcal{O}(64^2)$ token-pair complexity, while node projection, tokenization, and scatter retain dependence on N (approximately $\mathcal{O}(NM)$ for the assignment operations). Thus, token attention avoids an $\mathcal{O}(N^2)$ attention matrix but the full operator is not independent of mesh size. The MLP ratio is 4, giving hidden dimension 1,024.

4) Scatter Decoder: Predictions are projected back to the full mesh using the *same* soft-assignment weights:

$$\hat{\boldsymbol{f}}_n = \sum_{m=1}^{M} \alpha_{nm}\, \boldsymbol{\tau}_m^{(\text{out})} \in \mathbb{R}^d \tag{23}$$

Per-field linear projections ($256 \to 1$) with bias terms produce the final output $\hat{\boldsymbol{S}} \in \mathbb{R}^{B\times 4\times N}$.

5) Architecture Parameters: Direct model instantiation gives 3,244,872 trainable parameters: 66,816 in the branch encoder, 17,472 in mesh embedding/tokenization, 3,159,040 in four transformer blocks, and 1,544 in normalization/output terms.

C. *Tier 2: Clifford-Algebra Operator*

Fluid dynamics motivates rotation-aware representations: if geometry and vector fields are rotated consistently, scalar observables remain invariant. Standard MLPs do not impose this structure. The studied Clifford operator [6] uses the Clifford algebra Cl(3,0) as a grade-structured inductive bias, without claiming exact end-to-end equivariance for the implemented dense readout.

1) Multivector Representation: In Cl(3,0), every element is a multivector with 8 independent grades:

TABLE III
CL(3,0) MULTIVECTOR GRADE DECOMPOSITION

Grade	Basis	Physical Meaning	Index
0 (scalar)	1	Pressure, temperature	0
1 (vector)	e_1, e_2, e_3	Velocity components	1–3
2 (bivector)	e_{12}, e_{13}, e_{23}	Vorticity	4–6
3 (trivector)	e_{123}	Volume element	7

The metric signature is $(+,+,+)$: $e_i^2 = +1$, $e_i e_j = -e_j e_i$ for $i \neq j$. The geometric product $\boldsymbol{a} \otimes \boldsymbol{b}$ simultaneously computes inner products (grade-lowering) and outer products (grade-raising), naturally coupling scalar and vector physics. We implement the full 8×8 bilinear product table as a differentiable operation.

2) CliffordLinear Layer: The core operation replaces standard matrix multiplication with the geometric product:

$$\boldsymbol{h}_j^{(\text{out})} = \sum_{i=1}^{C_{\text{in}}} \boldsymbol{W}_{ji} \otimes \boldsymbol{h}_i^{(\text{in})} + \boldsymbol{b}_j \tag{24}$$

where $\boldsymbol{W}_{ji} \in \mathbb{R}^8$ are learned multivector weights and $\otimes$ is the Cl(3,0) geometric product. Each output channel is a sum of geometric products across all input channels—the algebraic generalization of a standard linear layer.

3) Scalar-Gate Residual: To retain grade structure within each residual block, only the grade-0 (scalar) component is passed through a nonlinearity:

$$\boldsymbol{h} \leftarrow \text{CliffordLinear}(\boldsymbol{h}) \tag{25}$$

$$\boldsymbol{h} \leftarrow \text{LayerNorm}(\boldsymbol{h}) + \boldsymbol{h}_{\text{res}} \tag{26}$$

$$\boldsymbol{h}_{[\ldots,0]} \leftarrow \text{GELU}(\boldsymbol{h}_{[\ldots,0]}) \tag{27}$$

This block-level design is symmetry-motivated, but exact network-level equivariance also depends on the embedding, normalization, and readout.

4) Input/Output Projection: The `PhysicsToMultivector` module embeds branch parameters (scalars $\to$ grade-0) and mesh coordinates (vectors $\to$ grade-1) into a $C = 32$-channel multivector representation: $[B, N, 32, 8]$. After four Clifford layers, `MultivectorToFields` actually flattens the $32 \times 8 = 256$ components and applies a generic $256 \to 4$ dense layer. This can mix grades; velocity magnitude is emitted directly as one scalar channel rather than computed as a grade-1 norm.

5) Scope of the Equivariance Claim: For a grade-preserving Clifford network, an equivariance target would be

$$\mathcal{G}_{\text{Clifford}}(\boldsymbol{R}\,\boldsymbol{y}, \boldsymbol{u}) = \boldsymbol{R}_{\text{applied}} \cdot \mathcal{G}_{\text{Clifford}}(\boldsymbol{y}, \boldsymbol{u}) \tag{28}$$

but Eq. (28) is not guaranteed by the implemented dense readout and has not been verified by rotated-input tests. Accordingly, "Clifford" here describes the internal multivector operations, not a proven symmetry of the full input–output map. A grade-selective decoder and numerical rotation-consistency study are required to evaluate end-to-end equivariance.

6) Parameter Efficiency: With $C = 32$ channels and four layers, direct model instantiation gives 37,380 trainable parameters, 97.4% fewer than DeepONetFourier (approximately 38.8× smaller). This remains substantial parameter efficiency, but accuracy and data-efficiency advantages require controlled evaluation rather than inference from size alone.

VI. LOCA INDICATOR DETECTION PIPELINE

A. *Feature Translation*

Raw synthetic-field predictions are translated into eleven scalar features—average pressure, pressure gradient, pressure drop, mass flow rate, inlet velocity, maximum and average turbulence, average temperature, temperature difference, and velocity/pressure standard deviations—which are then mapped through a blended severity model to generate seven NPPAD-like signals.

1) Blended Severity Model: The implemented effective severity combines a break-size sigmoid with a clipped field-anomaly heuristic:

$$\begin{aligned} \eta_{\text{input}} &= \left[1 + \exp(-0.8(b-5))\right]^{-1}, \\ \eta_{\text{eff}} &= 0.8\,\eta_{\text{input}} + 0.2\,\min(\eta_{\text{field}}, 1). \end{aligned} \tag{29}$$

where b is expressed in percent and η_{field} averages thresholded turbulence, pressure-variation, and inlet-flow-deficit signals from the neural-operator predictions. These thresholds and the subsequent NPPAD-like transformations are heuristic and have not been fitted to paired field/system measurements. The 80/20 blend means the classifier output is substantially driven by prescribed break size; see Section IX.

The seven NPPAD-mapped signals (Table IV) are computed via deterministic proxy transformations parameterized by η_{eff}. The ranges below reflect qualitative trend directions of mapped indicators (not raw plant measurements) under the synthetic benchmark:

TABLE IV
NPPAD-MAPPED INDICATORS: QUALITATIVE TREND UNDER INCREASING LOCA SEVERITY (MAPPED PROXY VALUES FROM SYNTHETIC BENCHMARK; NOT RAW PLANT MEASUREMENTS.)

Signal	Unit	Trend: Normal $\rightarrow$ LOCA
Primary Pressure (P)	bar	Decreases
Avg. Coolant Temp. (TAVG)	°C	Decreases
RCS Flow (WRCA)	kg/s	Decreases
SG-A Pressure (PSGA)	bar	Varies
Subcooling Margin (SCMA)	°C	Decreases
Boiling-margin proxy	—	Increases on mapped proxy scale
Hot-Cold Leg ΔT	°C	Decreases

B. Gradient Boosting LOCA Indicator Classifier

A Gradient Boosting Classifier (200 trees, depth 4, learning rate 0.05, min. 20 samples/leaf) operates on the seven scaled NPPAD-mapped signals. Training data contain 302 Normal and 45,176 LOCA rows from the PCTRAN-simulated NPPAD dataset [17], plus 3,800 synthetic transitional rows (49,278 total before the 80/20 split). The classifier outputs a score $S_{\text{LOCA}} \in [0, 1]$; the decision threshold is 0.5. No calibration study is available, so S_{LOCA} is not interpreted as a calibrated probability.

Because η_{eff} derives 80% of its signal from the prescribed break-size parameter, the main classifier result is an indicator-translation demonstration rather than independent end-to-end validation. Table XI reports the available five-seed controlled translation-path ablation. The "direct term removed" variant removes the explicit $0.8\,\eta_{\text{input}}$ term from feature translation, but break size remains an operator branch input. More importantly, the evaluation script samples $b \in [0, 0.5]$ for Normal rows and $b \in [1, 10]$ for LOCA rows according to the known label. The ablation therefore tests whether field translation preserves a label-conditioned severity signal; it is not leakage-free or field-independent detection evidence.

A subsequent validation study should sample operating conditions independently of the class label, preserve scenario-level group independence, and report threshold-dependent probability-of-detection and false-alarm operating curves with uncertainty.

VII. Implementation Details

A. Software and Hardware

The complete pipeline is implemented in Python 3.8+ using PyTorch (≥2.0), scikit-learn, NumPy, and Matplotlib. Archived configurations target a single NVIDIA RTX 4060 GPU (8 GB VRAM) and CUDA AMP. The available result artifacts do not include a complete per-run software/hardware manifest, which is required in the planned reproducibility archive.

B. Training Protocol

All neural operators share a common training framework:

TABLE V
TRAINING HYPERPARAMETERS (STANDARD DEEPONETFOURIER / TRANSOLVER / CLIFFORD PATHS; REPORTED SEED-42 TRANSOLVER SWEEP EXCEPTIONS NOTED)

Hyperparameter	DeepONet-F	Transolver	Clifford
Optimizer	Adam	Adam	Adam
Learning rate	10^{-3}	10^{-3}	10^{-3}
LR schedule	ReduceOnPlateau	CosineAnnealing	CosineAnnealing
patience/period	20 epochs	—	—
factor	0.5	—	—
Batch size	4	16 standard; 4 in reported sweep	16
Max epochs	2000	500 standard; 2000 in reported sweep	500
Early stopping	50 epochs	50 epochs	50 epochs
Mixed precision	Yes	Yes	Yes
Dropout	5%	10%	—
Loss	MSE + index proxies	MSE	MSE
Gradient clip	1.0	1.0	—

C. Unified Architecture Selection

The pipeline supports seamless switching via a `model_config.yaml` configuration:

- `model_version: deeponet_fourier` — Tier 1 default
- `model_version: transolver` — Tier 2 transformer
- `model_version: clifford` — Tier 2 geometric algebra

All operators accept the same input tensors ($[\mathtt{B}, 3]$ parameters, $[\mathtt{N}, 3]$ mesh coordinates) and produce $[\mathtt{B}, 4, \mathtt{N}]$ field predictions, enabling drop-in replacement.

VIII. Experimental Results

Result provenance and stability: Table VII gives exact pooled metrics recomputed from `deeponet_fourier_best.pth` over all 300 held-out analytic synthetic cases. Table VIII reports the separately archived seed-42 Transolver-inspired metric record over the same full test set; this MSE-only run used a 2,000-epoch budget and retained the minimum-validation-loss state from epoch 1,864. Table X gives exact values stored with the single classifier artifact (seed 42). Table XI is a five-seed analysis, subject to its class-conditioned-input limitation. Because a five-seed operator sweep has not been completed, cross-architecture accuracy is not reported as a stable comparative result; a matched-budget, multi-seed study is specified in Section IX-E.

A. DeepONetFourier Training Convergence

Table VI preserves a representative archived training trace: training began at $\mathcal{L}_{\text{train}} = 0.815$ and the logged validation objective reached 9.11×10^{-4}. Because the trace and the separately archived checkpoint used for Table VII do not share a complete immutable run manifest, they are not asserted to be the same run.

TABLE VI
REPRESENTATIVE ARCHIVED DEEPONETFOURIER TRAINING TRACE (SELECTED EPOCHS; NOT USED AS THE TEST-METRIC RECORD)

Epoch	Train Loss	Val Loss	LR
1	0.81485	0.03887	1.0×10^{-3}
5	0.01694	0.00480	1.0×10^{-3}
10	0.00861	0.00244	1.0×10^{-3}
20	0.00496	0.00197	1.0×10^{-3}
50	0.00389	0.00131	1.0×10^{-3}
80	0.00321	0.00103	5.0×10^{-4}
120	0.00298	9.72×10^{-4}	5.0×10^{-4}
160	0.00288	9.28×10^{-4}	2.5×10^{-4}
200	0.00284	9.15×10^{-4}	1.25×10^{-4}
305	0.00281	9.11×10^{-4}	1.25×10^{-4}

B. Synthetic-Benchmark Field Reconstruction Accuracy

Figs. 2–5 present comparative reconstructions for the same held-out synthetic case. All panels were regenerated from the archived checkpoints as 300-dpi lossless PNGs. Within each field, ground truth and every architecture use identical field levels, and all absolute-error panels use one common error scale. These figures are qualitative single-case diagnostics; full-test metrics are available for DeepONetFourier and the separate seed-42 Transolver-inspired run in Tables VII and VIII, respectively. Clifford-algebra accuracy remains limited to the matched-scale case-0 panel.

Table VII summarizes DeepONetFourier field accuracy on the 300-sample test set under the synthetic benchmark.

TABLE VII
DEEPONETFOURIER POOLED FIELD ACCURACY (ALL 300 HELD-OUT SYNTHETIC CASES, SINGLE ARCHIVED CHECKPOINT)

Field	R^2	Rel. L_2	MAE (norm.)
Pressure	0.9961	0.0158	0.00947
Velocity magnitude	0.9958	0.0348	0.01056
Turbulence k	0.9951	0.0449	0.00666
Temperature	0.9962	0.0278	0.01221

Here, pooled $R^2 = 1 - \sum_i (y_i - \hat{y}_i)^2 / \sum_i (y_i - \bar{y})^2$, relative $L_2 = \|\hat{\boldsymbol{y}} - \boldsymbol{y}\|_2 / \|\boldsymbol{y}\|_2$, and MAE is computed after flattening all cases and nodes for each normalized field. Per-case R^2 means (population SD) were 0.9879 (0.0137), 0.9950 (0.0011), 0.9675 (0.0569), and 0.7355 (0.0126), respectively. Temperature's lower per-case R^2 reflects its small within-case spatial variance despite high pooled performance; reporting both scopes prevents metric-selection ambiguity.

TABLE VIII
SEED-42 TRANSOLVER-INSPIRED POOLED FIELD ACCURACY (ALL 300 HELD-OUT ANALYTIC SYNTHETIC CASES; MSE-ONLY OBJECTIVE)

Field	R^2	Rel. L_2	MAE (norm.)
Pressure	0.991775	0.022916	0.013808
Velocity magnitude	0.993163	0.044359	0.012909
Turbulence k	0.980636	0.089384	0.012279
Temperature	0.995113	0.031441	0.013671

The Transolver-inspired record has a mean across the four pooled field R^2 values of 0.990172 and a minimum validation MSE of 3.03474×10^{-4}. These values quantify one initialization on the fixed-mesh analytic benchmark; they neither establish run-to-run stability nor isolate architecture from the non-identical DeepONetFourier objective.

C. Neural Operator Comparison

Key observations under the present benchmark:

- **DeepONetFourier** reconstructs the four synthetic fields with pooled $R^2 = 0.9951$–0.9962 on one fixed-mesh checkpoint. Its different objective prevents attributing any comparative advantage to architecture alone.
- **The Transolver-inspired operator** supplies global token attention, has 3.245M parameters, and attained pooled $R^2 = 0.9806$–0.9951 across the four fields in the seed-42 full-test run.
- **The Clifford-algebra operator** is compact at 37,380 parameters. Those properties require dedicated learning-curve and rotation-consistency experiments.

D. LOCA Indicator Classification Performance

Fig. 6 shows the LOCA indicator detection performance panel, and Table X summarizes classifier metrics. As discussed in Section VI, the classifier output is substantially driven by the input break-size parameter via η_{eff}, so these results should be read as an indicator-translation demonstration rather than as end-to-end digital twin validation.

The archived classifier metrics do not retain a break-size-stratified error analysis, and no reliability diagram, Brier score, or expected calibration error was computed. Approximate score-versus-break-size values from the prior draft are therefore omitted rather than presented as calibration evidence. A reliability-oriented follow-up will use independent group-aware splits and report threshold-dependent probability-of-detection/false-alarm curves, proper scoring rules, and calibration intervals.

E. Runtime Scope

The inference implementation measures neural-operator execution separately and provides an end-to-end path comprising operator inference, feature translation, and classification for one case. Earlier project notes contained latency and CFD-speedup estimates, but no immutable timing log records the warm-up policy, device synchronization, repetitions, latency distribution, software stack, or a matched Fluent comparator. Those estimates are therefore excluded from the paper's quantitative findings. A future benchmark will archive these timing boundaries and report distributional latency on specified hardware alongside a measured CFD reference.

IX. DISCUSSION

A. Methodological Contributions

This work makes three primary methodological contributions. First, it demonstrates that three distinct neural-operator paradigms—enhanced branch–trunk decomposition,

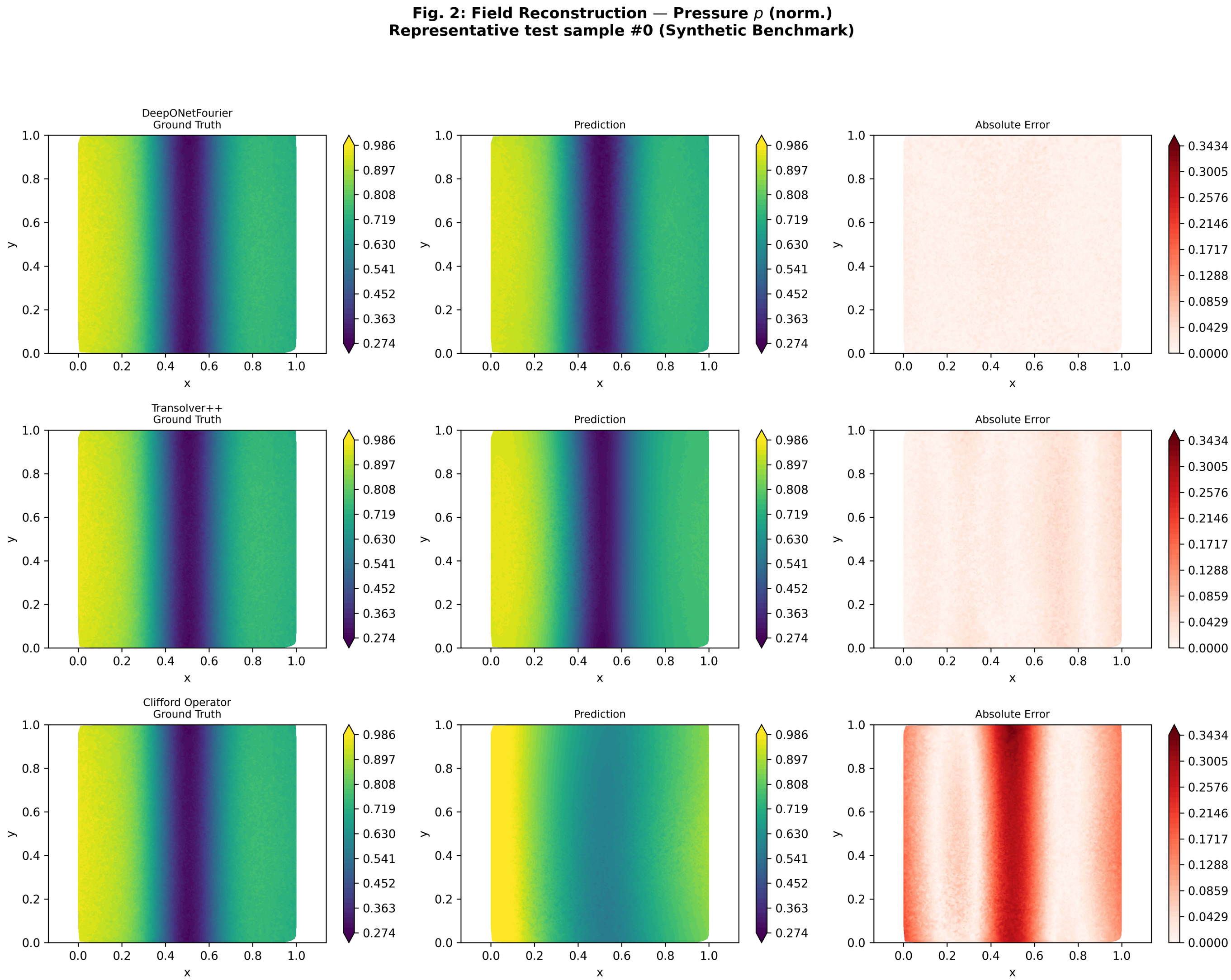

Fig. 2. Normalized pressure reconstruction for held-out synthetic case 0. Rows show DeepONetFourier, the Transolver-inspired operator, and the Clifford-algebra operator; columns show common-scale ground truth, common-scale prediction, and common-scale absolute error.

TABLE IX
NEURAL OPERATOR ARTIFACT COMPARISON (SYNTHETIC BENCHMARK; NON-IDENTICAL LOSS SETTINGS)

Metric	**DeepONet-F**	**Transolver**	**Clifford**
Total parameters	1,451,012	3,244,872	37,380
Retained accuracy evidence	Pooled R^2: 0.9951–0.9962	Seed-42 pooled R^2: 0.9806–0.9951	Matched-scale case-0 panel
Output shape	$[B, 4, N]$	$[B, 4, N]$	$[B, 4, N]$
Training objective	MSE + index proxies	MSE	MSE
Evaluated geometry	Fixed shared mesh	Fixed shared mesh	Fixed shared mesh

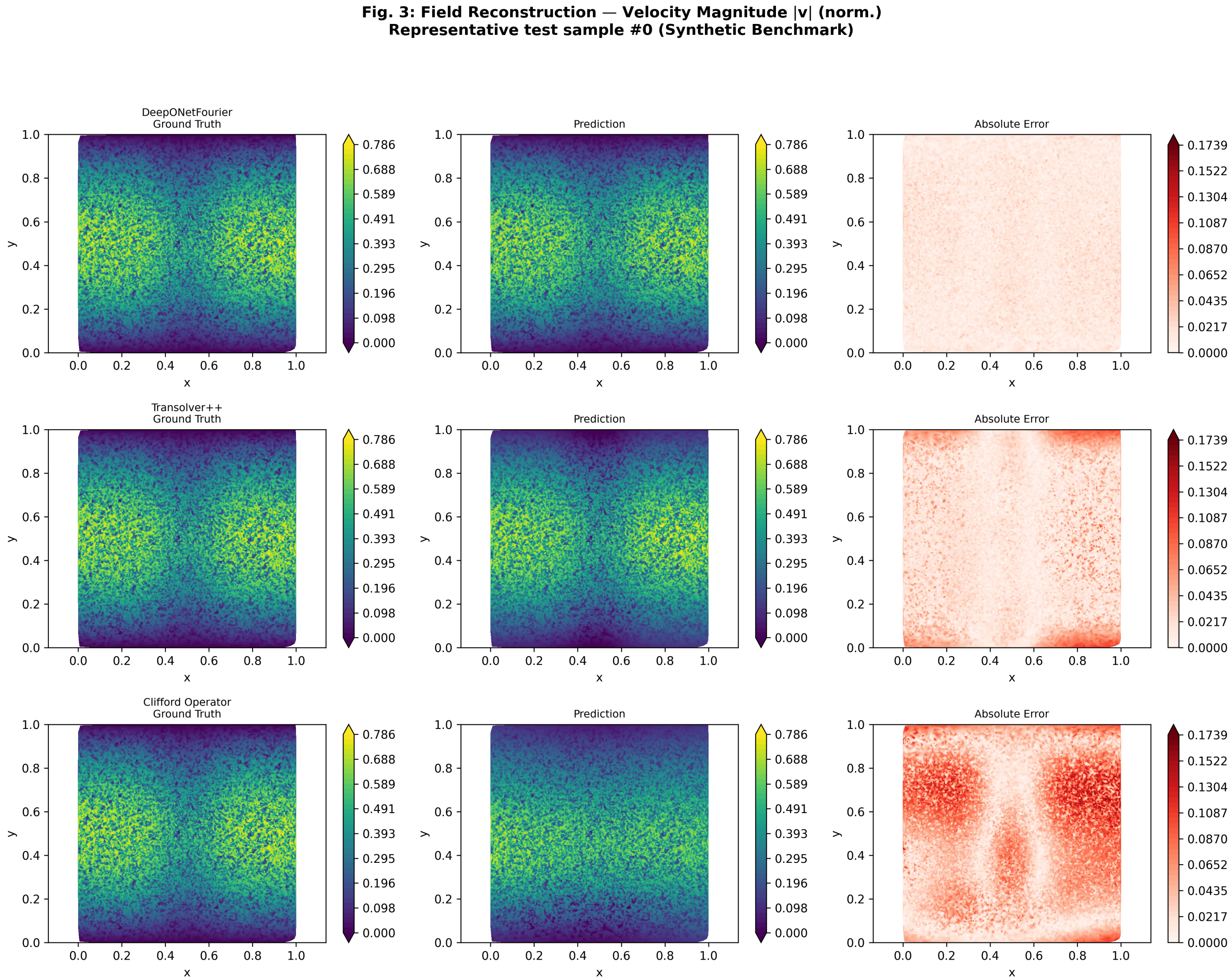

Fig. 3. Normalized velocity-magnitude reconstruction for held-out synthetic case 0, using the matched layout and scales described in Fig. 2.

TABLE X
LOCA INDICATOR CLASSIFIER PERFORMANCE (NPPAD-SIMULATED + SYNTHETIC TRANSITION DATA, ONE SEED-42 SPLIT)

Metric	Score
Accuracy	0.977374
Precision	0.987281
Recall (Sensitivity)	0.989063
F1 Score	0.988172
ROC-AUC	0.990554

TABLE XI
FIVE-SEED TRANSLATION-PATH ABLATION (MEAN ± POPULATION SD). THE DIRECT SEVERITY TERM IS REMOVED IN THE SECOND COLUMN, BUT LABEL-CONDITIONED BREAK SIZE REMAINS AN OPERATOR INPUT.

Metric	80/20 Blend	Direct Term Removed
ROC-AUC	0.999997 ± 0.000003	0.999977 ± 0.000046
Recall	1.000000 ± 0.000000	1.000000 ± 0.000000
F1	0.999956 ± 0.000041	0.999989 ± 0.000022
Accuracy	0.999912 ± 0.000082	0.999978 ± 0.000044

mesh-token attention, and geometric-algebra encoding—can share one surrogate-to-indicator pipeline. Second, it establishes exact synthetic-test metrics for the archived DeepONet-Fourier checkpoint, a separately executed seed-42 Transolver-inspired run, and a dataset-matched unmodified DeepONet baseline while documenting the non-identical objectives. Code inspection identifies two index-space proxy terms, preventing a conservation claim and motivating a geometry-aware retraining experiment. Third, the work exposes the current validation boundary: analytic fields, PCTRAN-simulated classifier data, label-conditioned translation, single-run operator results, and the absence of a matched CFD timing reference.

B. Architectural Trade-Off Analysis

The three architectures address different deployment priorities and the following observations are framed as *preliminary inductive-bias findings* under non-identical optimization settings:

DeepONetFourier has 54.1% fewer parameters than the retrained DeepONet baseline and achieved pooled R^2 =

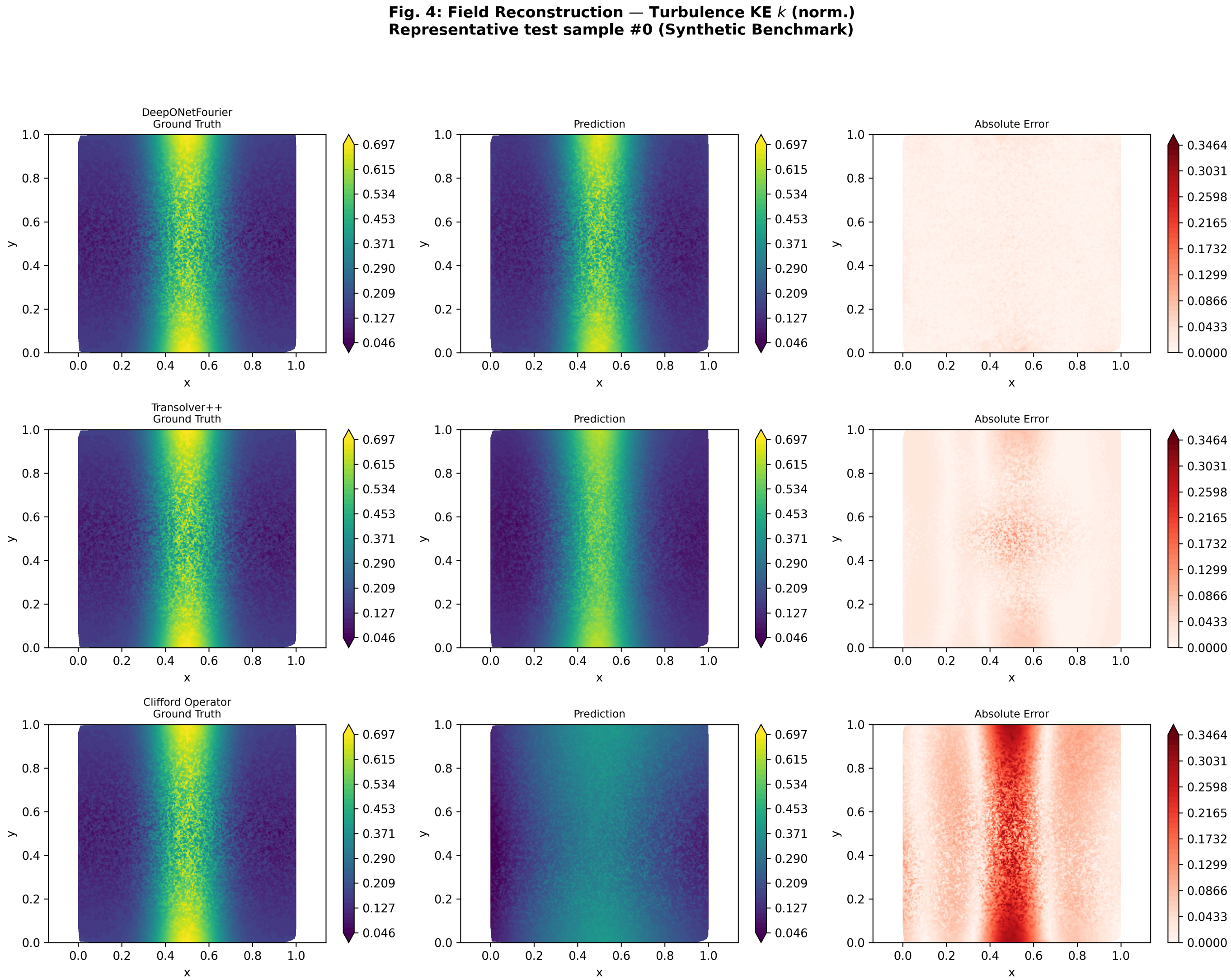

Fig. 4. Normalized turbulent-kinetic-energy reconstruction for held-out synthetic case 0, using the matched layout and scales described in Fig. 2.

0.9951–0.9962 on the 300-case full test set. Fourier encoding is an explicit high-frequency representation mechanism; the incremental effects of that encoding, adaptive activation, and the two proxy terms have not been isolated under matched retraining.

The Transolver-inspired operator provides global interactions over $M = 64$ tokens, with $\mathcal{O}(M^2)$ token-attention complexity, and is the largest studied model at 3.245M parameters. Its seed-42 full-test record attained a mean field R^2 of 0.990172, with all four pooled field values above 0.98. Because this is a single MSE-only run and all training and evaluation use the same mesh, run-to-run stability and transfer across topology remain untested.

The Clifford-algebra operator is the most compact studied model at 37,380 parameters. Its internal Cl(3,0) operations preserve a useful grade structure, but the dense $256 \to 4$ decoder mixes grades. Learning-curve and rotation-consistency experiments are required before either advantage can be asserted.

C. Surrogate-Monitoring Utility and Limits of Current Validation

The current results represent upper-bound methodological performance under controlled, physics-inspired analytic synthetic conditions. Several important caveats apply:

The synthetic generator encodes qualitative pipe-flow patterns but does not solve governing equations or establish conservation consistency. The reported DeepONetFourier and Transolver-inspired pooled R^2 values therefore quantify interpolation on that analytic generator's distribution, not accuracy relative to Fluent CFD, experiments, or plant transients.

The blended severity model ($\eta_{\text{eff}} = 0.8\eta_{\text{input}} + 0.2\min(\eta_{\text{field}}, 1)$) means the main LOCA score is substantially driven by prescribed break size. The available five-seed ablation removes the explicit translator term, but its break-size input is sampled conditionally on the class. A label-independent evaluation remains necessary to quantify whether reconstructed fields add detection information.

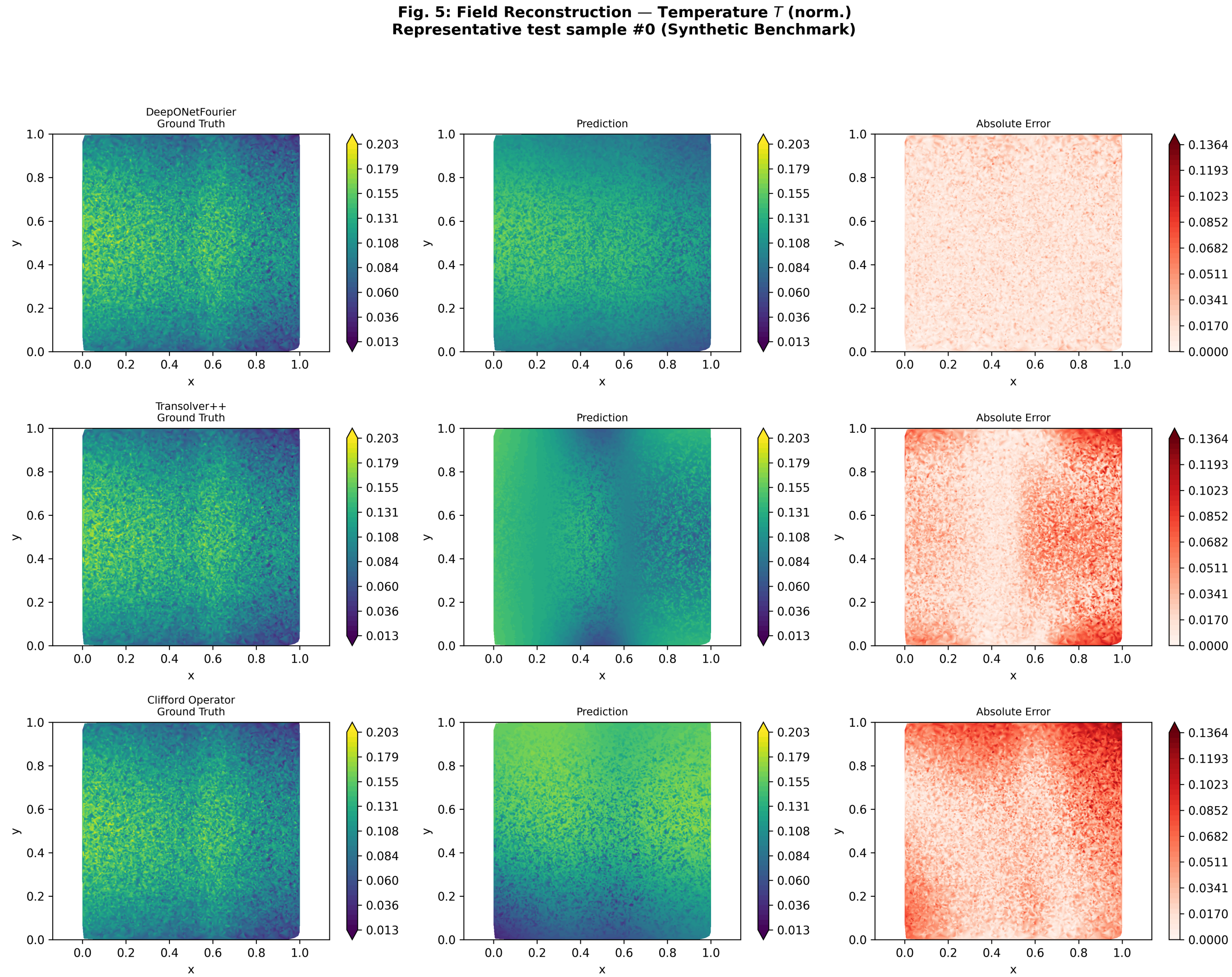

Fig. 5. Normalized temperature reconstruction for held-out synthetic case 0, using the matched layout and scales described in Fig. 2.

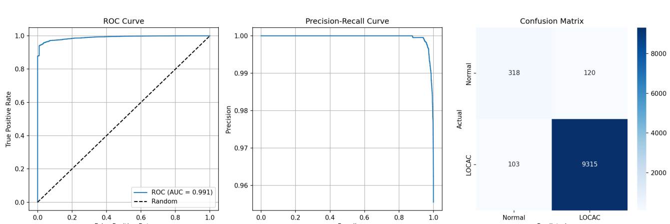

Fig. 6. Classifier ROC curve (AUC 0.990554), precision–recall curve, and confusion matrix for one seed-42 split of PCTRAN-simulated NPPAD rows plus synthetic transition augmentation. This is classifier-level evidence, not independent end-to-end digital-twin validation.

D. Path to Full-Fidelity Validation

The path to full-fidelity validation is direct but unfinished: repair and verify the Fluent automation on the intended elbow geometry; generate an independently quality-controlled case matrix spanning break severity, inlet velocity, and temperature; fit all scalers on training cases only; reserve grouped geometry and operating-condition holdouts; and retrain the operators under matched seeds and losses. The repository contains an unevaluated Fluent journal-generation and CSV-preprocessing prototype, whereas the present benchmark uses a separate straight cylindrical proxy. No complete Fluent case matrix or corresponding evaluated operator checkpoint was available for this study.

E. Limitations and Future Work

Missing baselines: A dataset-matched unmodified DeepONet result is now included, but plain MLP and FNO baselines remain absent. A component ablation is also required to isolate Fourier features, adaptive activations, and each proxy term.

Run-to-run stability: The reported DeepONetFourier and Transolver-inspired field results and the primary classifier artifact are single-run; Clifford-algebra field accuracy has not been established over the complete test set. Only the constrained translation-path ablation has five seeds. Future comparisons will use predeclared seeds, immutable run manifests, and mean $\pm$ sample standard deviation under both current and matched-objective protocols.

Split and preprocessing protocol: The current field split samples randomly from a regular parameter grid and therefore tests in-distribution interpolation rather than extrapolation to a held-out operating regime. In addition, the current preprocessor fits min–max scalers before partitioning, so held-out extrema influence scaling. The classifier uses a row-level split that can place time steps from one simulated transient in both training and test sets. Future validation will fit preprocessing on training data only and use parameter-blocked field splits and transient-grouped classifier splits.

Non-identical optimization: DeepONetFourier uses a doubled value-fidelity term plus index-gradient and velocity-variation proxies, whereas the other operators use MSE only. A controlled comparison requires matched objectives and tuning budgets; present cross-architecture figures are qualitative artifact inspections, not a ranking.

Steady-state assumption: LOCA events are transient, and the present results address only steady-state field reconstruction. No enabled configuration, checkpoint, or transient result was found for the prototype temporal module; temporal operator learning is therefore future work. Evaluation should use transient CFD and independently aligned sensor sequences and report detection delay, probability of detection, false alarms, and distribution-shift performance.

Uncertainty quantification: The evaluated pipeline produces point predictions and classifier scores only. Diffusion is disabled by default; no trained diffusion checkpoint, held-out sampling protocol, coverage result, or decision-level propagation artifact was found. DDPM-based turbulence sampling is therefore future work, and a classifier score is not predictive uncertainty. A complete study must report nominal versus empirical coverage, interval width, proper scores, and calibration and propagate uncertainty through feature translation and classification.

Detection-threshold and false-alarm analysis: The classifier currently reports performance at a single decision threshold (0.5) via aggregate metrics (ROC-AUC, recall, F1). A reliability-oriented treatment should additionally report the detection-threshold/false-alarm-rate trade-off explicitly (e.g., an operating-point analysis tied to an acceptable false-alarm budget for operator-facing screening), since false negatives and false positives carry very different consequences in a LOCA-screening context.

AI-model failure modes: This study does not analyze distribution shift between analytic synthetic fields and Fluent/experimental data, robustness to noise or simulator shift in the PCTRAN-derived features, or behavior outside $v \in [4.0, 6.0]$ m/s, $b \in [0, 10]\%$, and $T \in [290, 320]°$C. Such evidence is central to a reliability argument [20], [21].

Multi-fidelity learning: A residual multi-fidelity module combining coarse-RANS and fine-LES predictions ($u_{\text{final}} = u_{\text{base}} + u_{\text{residual}}$) is a planned extension; not evaluated in the present results.

Wall shear stress: A post-processing module for $\tau_w = \mu\,\partial u/\partial y$ with corrosion-risk assessment is a planned extension for pipe-integrity monitoring; not evaluated in the present results.

X. Conclusion

This paper has presented a proof-of-concept, multi-architecture neural-operator workflow for Cold-Leg LOCA screening in the AP1000 PWR, comparing three operator paradigms and coupling the archived DeepONetFourier surrogate to a downstream classification pipeline. The principal findings were obtained under the physics-inspired analytic synthetic benchmark described in Section III:

- **DeepONetFourier** has 1,451,012 parameters, 54.1% fewer than the 3,162,116-parameter retrained DeepONet baseline. Its archived checkpoint attains pooled test $R^2 = 0.9951$–0.9962 across the four analytic synthetic fields. Fourier coordinates and adaptive activations are combined with two explicitly scoped index-space regularizers.
- **The Transolver-inspired operator** has 3,244,872 parameters and provides global attention over 64 learned mesh tokens. Its separately archived seed-42 MSE-only run attained pooled test $R^2 = 0.9806$–0.9951 across the four analytic synthetic fields (mean 0.990172); a completed multi-seed study is still required for a stable comparison.
- **The Clifford-algebra operator** is the most compact studied architecture at 37,380 parameters, approximately 38.8× fewer than DeepONetFourier. Its internal multivector operations provide a grade-structured inductive bias; rotation-aware decoding remains to be evaluated.
- The single-split classifier artifact reaches ROC-AUC 0.990554, recall 0.989063, and F1 0.988172 on PCTRAN-simulated NPPAD rows plus synthetic transition augmentation. The five-seed translation-path ablation remains label-conditioned, so neither result establishes independent LOCA detection from predicted fields.
- The implementation provides a unified single-case inference and timing path for all three architectures. Quantitative speedup is reserved for a matched benchmark with archived timing distributions and a measured CFD reference.

Collectively, these results demonstrate methodological feasibility rather than validated screening capability. Full-fidelity CFD/experimental validation, geometry-aware physical derivatives, label-independent detection, matched-loss multi-seed evaluation, calibrated uncertainty, and a reliability safety case are prerequisites for any safety-relevant claim. The shared configuration interface makes those experiments tractable, but software integration alone is not evidence of operational readiness.

Data Availability Statement

Project source code, including the analytic synthetic-field generator, is publicly available at https://github.com/Punyansh26/Digital-Twin-for-NPP-Cold-Leg-LOCAC-detection. The exact local split, checkpoints, metric JSON files, and revised figures used for this manuscript are available from the corresponding author upon reasonable request. The PCTRAN-simulated NPPAD collection is openly available under the Creative Commons Attribution 4.0 license from the originating publication and Figshare record [17]; DOI: https://doi.org/10.6084/m9.figshare.c.6238473.

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REFERENCES

[1] L. Lu, P. Jin, G. Pang, Z. Zhang, and G. E. Karniadakis, "Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators," *Nature Machine Intelligence*, vol. 3, pp. 218–229, 2021.

[2] T. Chen and H. Chen, "Universal approximation to nonlinear operators by neural networks with arbitrary activation functions and its application to dynamical systems," *IEEE Trans. Neural Networks*, vol. 6, no. 4, pp. 911–917, 1995.

[3] L. Lu, X. Meng, S. Cai, Z. Mao, S. Goswami, Z. Zhang, and G. E. Karniadakis, "A comprehensive and fair comparison of two neural operators (with practical extensions)," *Comput. Methods Appl. Mech. Engrg.*, vol. 393, p. 114778, 2022.

[4] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar, "Fourier neural operator for parametric partial differential equations," in *Proc. ICLR*, 2021.

[5] H. Wu, H. Luo, H. Wang, J. Wang, and M. Long, "Transolver: A fast transformer solver for PDEs on general geometries," in *Proc. ICML*, 2024.

[6] J. Brandstetter, R. van den Berg, M. Welling, and J. K. Gupta, "Clifford neural layers for PDE modeling," in *Proc. ICLR*, 2023.

[7] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville, "On the spectral bias of neural networks," in *Proc. ICML*, pp. 5301–5310, 2019.

[8] M. Tancik et al., "Fourier features let networks learn high frequency functions in low dimensional domains," in *Proc. NeurIPS*, 2020.

[9] H. Son, J. W. Jang, W. J. Han, and H. J. Hwang, "Sobolev training for physics-informed neural networks," *arXiv:2101.08932*, 2021.

[10] Westinghouse Electric Company, "AP1000 Design Control Document (DCD)," NRC ADAMS ML11171A500, Rev. 19, 2011.

[11] U.S. Nuclear Regulatory Commission, "Best-estimate calculations of emergency core cooling system performance," Regulatory Guide 1.157, May 1989.

[12] M. Grieves and J. Vickers, "Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems," in *Transdiscipl. Perspectives Complex Syst.*, Springer, 2017, pp. 85–113.

[13] Q. Huang, S. Peng, J. Deng, H. Zeng, Z. Zhang, Y. Liu, and P. Yuan, "A review of the application of artificial intelligence to nuclear reactors: Where we are and what's next," *Heliyon*, vol. 9, no. 3, Art. no. e13883, 2023, doi: 10.1016/j.heliyon.2023.e13883.

[14] Z. Hao et al., "GNOT: A general neural operator transformer for operator learning," in *Proc. ICML*, 2023.

[15] R. Hossain, F. Ahmed, K. Kobayashi, S. Koric, D. Abueidda, and S. B. Alam, "Virtual sensing-enabled digital twin framework for real-time monitoring of nuclear systems leveraging deep neural operators," *npj Materials Degradation*, vol. 9, Art. no. 21, 2025, doi: 10.1038/s41529-025-00557-y.

[16] M. Lee, S. Oh, C. Song, B. Cho, S. Baral, S. Khanal, M. Song, and J. Jeon, "Neural operator-based surrogate model for CFD: Helical coil steam generator in small modular reactor," *arXiv:2605.30277*, 2026, doi: 10.48550/arXiv.2605.30277.

[17] B. Qi, X. Xiao, J. Liang, L.-C. Po, L. Zhang, and J. Tong, "An open time-series simulated dataset covering various accidents for nuclear power plants," *Scientific Data*, vol. 9, Art. no. 766, 2022, doi: 10.1038/s41597-022-01879-1.

[18] R. Isermann, "Model-based fault-detection and diagnosis—status and applications," *Annual Reviews in Control*, vol. 29, no. 1, pp. 71–85, 2005, doi: 10.1016/j.arcontrol.2004.12.002.

[19] *IEEE Standard for System, Software, and Hardware Verification and Validation*, IEEE Std 1012-2016, 2017.

[20] Z. Ma, H. Bao, S. Zhang, M. Xian, and A. Mack, "Exploring advanced computational tools and techniques with artificial intelligence and machine learning in operating nuclear plants," NUREG/CR-7294, INL/EXT-21-61117, U.S. Nuclear Regulatory Commission, Feb. 2022. [Online]. Available: https://www.nrc.gov/reading-rm/doc-collections/nuregs/contract/cr7294/index

[21] Canadian Nuclear Safety Commission, U.K. Office for Nuclear Regulation, and U.S. Nuclear Regulatory Commission, "Considerations for developing artificial intelligence systems in nuclear applications," Aug. 2024. [Online]. Available: https://onr.org.uk/media/03zl1osf/canukus_trilateral_ai_principles_paper_2024_08_28-final.pdf

[22] A. Petruzzi and F. D'Auria, "Thermal-hydraulic system codes in nuclear reactor safety and qualification procedures," *Science and Technology of Nuclear Installations*, vol. 2008, Art. no. 460795, 2008, doi: 10.1155/2008/460795.

[23] U.S. Nuclear Regulatory Commission, "Computer codes," [Online]. Available: https://www.nrc.gov/about-nrc/regulatory/research/safety-codes.html. Accessed: Aug. 31, 2026.